

PRD: PARL-Explorer Pro (The "Bio-God" Edition)

1. Project Vision

To create a world-class, bilingual bioinformatics platform that proves the regulatory axis between **PARL** and **MAVS** in Colorectal Cancer (CRC). This tool bridges the gap between raw genomic data and high-impact publication figures.

2. Advanced Feature Architecture

2.1 Input Sources (The "Omics" Layer)

- **TCGA-COAD/READ:** Automated pipeline to fetch RNA-Seq and Clinical metadata.
- **GEO Integration:** Support for GSE datasets (e.g., GSE39582) to validate MAVS downregulation in independent cohorts.
- **Custom Upload:** CSV/TSV support for lab-specific experiments (qPCR or Western Blot data).

2.2 Analysis Tools (The "Logic" Layer)

- **MAVS Efficacy Proof:** Differential expression analysis specifically showing MAVS levels across CRC stages (I-IV) vs. Normal tissue.
- **Survival Correlation:** Multi-gene Kaplan-Meier analysis (e.g., "How do patients fare when BOTH PARL and MAVS are high?").
- **PPI & Network Analysis:** Integration with **STRING-DB** API to visualize physical interactions between PARL, MAVS, and downstream effectors (like RIG-I or IRF3).
- **GSEA & Pathway Mapping:** Custom "Mitochondrial Health" enrichment score using KEGG/GO.

2.3 AI & Predictive Tools

- **Substrate Cleavage Prediction:** Use sequence-based motifs to predict if PARL can physically cleave MAVS.
- **Literature Miner:** Use Gemini/LLM APIs to find "hidden" mentions of PARL/MAVS interactions in non-CRC papers to build the "novelty" argument for her paper.

2.4 Output (The "Publication" Layer)

- **Vector Graphics:** Export high-res (300DPI) SVG/PDF plots ready for journal submission.

- **Bilingual Toggle:** Real-time UI translation (EN/ZH) for global collaboration.

3. Recommended Open-Source Toolkit

We will "re-skin" and wrap these gold-standard tools to make them beautiful:

- **ComplexHeatmap (R/Python):** For the expression patterns.
- **Cytoscape.js:** For the interactive Protein-Protein interaction (PPI) maps.
- **Gseapy:** For the enrichment analysis.
- **Streamlit-Echarts:** For high-performance, interactive front-end charts.

4. Technical Strategy

- **Frontend:** React.js with Tailwind CSS (for the "Dark Mode / Lab" aesthetic).
- **Backend:** Python FastAPI (to handle heavy bio-processing).
- **State:** Redux to manage the bilingual toggle and gene selection states.